

Detailed 8 x 8 Window Attention in U-MoE-Fusion

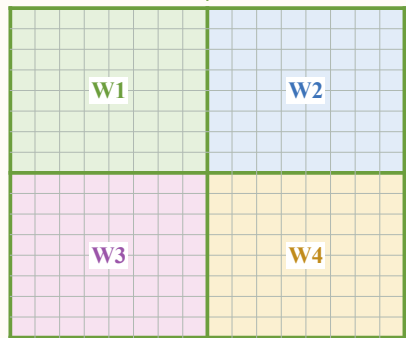
Regular Non-Shifted Window Attention Path

Spatial Windowing

Input Feature F $B \times H \times W \times 96$

Layer Normalization

Reflect Pad H and W to multiples of 8
training: 170 x 170 to 176 x 176



Flatten: $(B \times nW) \times 64 \times 96$
train $nW = 22 \times 22 = 484$
regular windows; shift size = 0

Q / K / V and 8 Heads

8 Head Outputs $(B \times nW) \times 8 \times 64 \times 12$

Fused Scaled Dot-Product Attention
 $\text{softmax}(Q K^T / \sqrt{12}) + B_{\text{rel}} V$

Q
 $8 \times 64 \times 12$

K
 $8 \times 64 \times 12$

V
 $8 \times 64 \times 12$

Linear QKV
96 to 288

Window Tokens
 64×96

Reverse and Residual

Head Outputs
 $8 \times 64 \times 12$

Concat Heads 64×96

Linear Projection 96 to 96

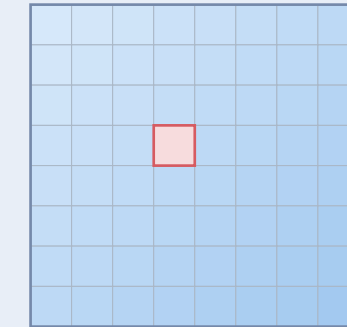
Reshape Each Output
64 tokens to 8×8

Window Reverse
and Crop to $H \times W$

Attention Residual
 $F_{\text{attn}} = F + \text{WindowAttn}(\text{LN}(F))$

Inside One 8 x 8 Window

Final W96L: window = 8, C = 96, heads = 8, d = 12

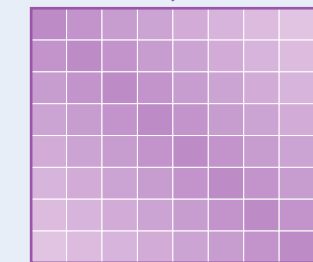


64 spatial tokens

Relative Position
Bias Table
 $15 \times 15 = 225$ / head

One Head
 Q_h, K_h, V_h
 64×12

$A_h = \text{softmax}(Q_h K_h^T / \sqrt{12}) + B_{\text{rel}}$
 $O_h = A_h V_h$



Head Output
 O_h 64×12

A_h 64×64

PyTorch fused SDPA mask = None no shifted windows

Complete Legend

Input / Residual

LayerNorm

Reflect Pad

8 x 8 Windows

Q / K / V

Relative Bias

Fused SDPA

Concat / Projection